

- 7 (a) (i) the half life / count rate / rate of decay / activity is the same no matter what external factors / environmental factors **or** two named factors such as temperature and pressure changes are applied B1 [1]
- (ii) the observations of the count rate / count rate / rate of decay / activity / radioactivity during decay shows variations / fluctuations B1 [1]

(b)

property	α -particle	β -particle	γ -radiation
charge	(+)2e	−e	0
mass	$4u$	$9.11 \times 10^{-31} \text{ kg}$	0
speed	0.01 to 0.1 c	up to 0.99 c	c

one mark for each correct line B3 [3]

- (c) collision with molecules B1
causes ionisation (of the molecule) / electron is removed B1 [2]